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Seeking the optimal posture of the seated lumbar spine

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Prolonged sitting with poor posture is associated with the development of lower back pain. Ergonomic texts for physiotherapists contain diverging and confusing views on recommended postures for the lumbar spine when seated that will promote postural health and optimal functioning of the lumbar spine. A review of the literature reveals that proponents of both the lordosed and kyphosed lumbar seated position use similar arguments with contradictory conclusions. The arguments of those advocating the kyphosed lumbar seated posture are, however, often anecdotal and unsubstantiated by research. This paper evaluates the conflicting views and concludes that the lordosed seated posture, regularly interspersed with movement, is the optimal sitting posture and assists in maintaining lumbar postural health and preventing low back pain.

INTRODUCTION

An important goal in physiotherapy is to foster health and prevent injury. This paper deals with the postural health of the lumbar spine and the positioning of the spine to promote optimal functioning in chair sitting. Two ways to address this goal are to educate healthy individuals and re-educate people with disorders related to poor sitting posture regarding the beneficial effects of optimal sitting postures on the spine. This is especially important for people with low back dysfunction or those with largely sedentary lifestyles.

To explore this topic, it is first necessary to clarify what is meant by posture, seated posture, and healthy sitting posture with definitions that relate to Western modes of sitting. ''Posture is a term that indicates the relative position of the body segments during rest or activity'' (Twomey & Taylor, 1987, p. 51). Sitting is a posture where weight is transferred to the seat of a chair through the ischial tuberosities and surrounding soft tissue of the buttocks and thighs. Depending on chair design, weight also is transferred through the back rest and the arm rests to the floor through the feet, and occasionally through the shins.

In the context of this paper, seated posture is defined according to the effect of sitting on the lumbar spine curvature. A flexed seated posture is one that reverses the individual's standing (normally lordotic) lumbar curve into a kyphotic position. A lordotic or extended seated posture is one that maintains the lumbar curve close to that of an individual's lordotic standing position.

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The Posture Committee of the American Academy of Orthopaedic Surgery (1947, cited in Cailliet, 1981, p. 23) defined healthy posture as "skeletal alignment refined as a relative arrangement of the parts of the body in a state of balance that protects the supporting structures of the body against injury or progressive deformity." In sitting, there is no one ideal posture, nor should one posture be sustained. Healthy sitting posture, therefore, is best thought of as an active, not a static phenomenon. In the resting position, between movements, healthy sitting posture can be thought of as occurring when unnecessary (static) muscle activity, ligamentous tension, intradiscal pressure, and zygapophysial joint forces are minimized, and when the body weight is distributed evenly through the ischial tuberosities and thighs to the seat and through the torso via the backrest.

Excluded from consideration in this paper are healthy seated postures for those with bony or soft tissue anomalies or diseases, particularly spondylolisthesis and canal stenosis. Here we describe the effects of seating on the lumbar spine; only brief consideration, where appropriate, is given to the cervical spine. To ensure adequate review of the literature, we have reviewed studies from many disciplines. In view of the lack of well-designed epidemiological clinical trials or systematic reviews investigating optimal seated posture, it has been necessary to rely largely on studies with lower levels of evidence (e.g., cadaveric studies).

ACHIEVING HEALTHY LUMBAR SITTING POSTURE

To understand the concept of healthy sitting posture it is necessary to consider the effects of sitting on the intervertebral disc (IVD), the zygapophysial joints, and the muscles and ligaments of the lumbar spine. It also is essential to consider the importance of movement, the behavior of the sitter and the chair's role in healthy sitting. These aspects are summarized in Table 1.

In summary, healthy seated postures are those that lower intradiscal pressure, minimize creep, minimize unnecessary myoelectrical activity in the posterior lumbar musculature and the least tension in posterior ligaments, and encourage load-sharing between the IVD and zygapophysial joints. As noted in Table 1, creep is defined as the phenomenon in which deformation of a tissue increases with time when the load is held constant (Frankel & Nordin, 1980). No one posture is correct, and movement is essential to facilitate nutrition and relieve muscle fatigue. Finally, the behavior of the sitter must be considered, in order to address user education.

Educating the sitter in healthy seated postures and optimizing chair use is an essential part of treatment for the sedentary worker with back pain. Although a lack of efficacy of education alone has been demonstrated by Daltroy et al (1997), such results do not relieve the physiotherapist from the onus of improving the standard of education and understanding, while recognizing the limitations. Further, the literature contains conflicting opinions as to the correct seated lumbar posture, creating difficulty for the physiotherapist wishing to educate a patient. Advice on the "correct" sitting posture has not always been substantiated by research, and conclusions from research have not always been logical. The question therefore remains: Which models of seated posture fulfill the criteria for seated postural health?

MODELS OF SEATED POSTURE

Historically, recommended sitting postures have included maintaining a right angled (or cubist) sitting position, lumbar kyphosis, lumbar lordosis, a reclined position of the backrest and seatpan and a forward tilt of the backrest and seatpan. These postures are discussed with a view to their impact on seated postural health.

Right angled (or cubist) sitting position

This seated position has been so named because the hips, trunk, knees, and ankles

Table 1
Effects of sitting, movement, and behavior on the spine

IVD and pressure	The IVD consists of the nucleus pulposus and the anulus fibrosis. Sitting compresses the IVD, creating hydrostatic pressure in the nucleus that is resisted by the anulus and adjoining vertebral bodies (Bogduk & Twomey, 1987). The amount of hydrostatic pressure within the nucleus is affected by the manner in which one sits, flexed or erect. An IVD flexed by 5.5° exhibits a nuclear hydrostatic pressure 50% greater than an IVD extended by 3° (Adams & Hutton, 1983, 1985).
IVD and nutrition	The disc is avascular and receives nutrition from osmosis and changing pressure within the disc, which alters the fluid content of the disc (Kraemer, Kolditz, & Gowin, 1985; Kramer, 1977; Nachemson & Elfström, 1970). Flexion increases intradiscal pressure and extrudes fluid and waste products (Adams & Hutton, 1983, 1985; Nachemson, 1963). Extension decreases intradiscal pressure and fluid and nutrients are imbibed. (Kramer, 1977; Nachemson & Elfström, 1970).
IVD and creep	Creep is the phenomenon in which deformation of a tissue increases with time when the load is held constant (Frankel & Nordin, 1980). Sustained intradiscal pressure above 0.3 Megapascals causes fluid loss from the anulus and nucleus of the disc (Nachemson & Elfström, 1970). This fluid loss is an underlying characteristic of creep during axial compression of the IVD (Bogduk & Twomey, 1987) and is evidenced by loss of height in the IVD space. Sustained sitting results in progressive creep with increased compressive force on the anulus and less shock absorption by the IVD (Adams, McNally, Chinn, & Dolan, 1994; Adams, McNally, & Dolan, 1996; Ohshima et al, 1989).
Muscles	Prolonged static back muscle activity increases IVD pressure, causes muscle fatigue, and creates waste product buildup (Andersson, Örtengren, Nachemson, & Elfström, 1974a; Floyd & Roberts, 1958). Sitting postures requiring minimal muscle activity are most beneficial for postural health.
Joints	Load sharing between the zygapophysial joints and IVD results in lowered IVD pressure, thus promoting a healthy sitting posture. Load sharing differs according to the degree or lordosis or kyphosis (Adams & Hutton, 1985). These differences are discussed more fully in the body of the paper.
Ligaments	Postural health is best served by positions that tense the posterior ligamentous system least. Differences in sitting posture alter posterior ligament tensioning with ramifications for potential disc damage (Hedman & Fernie, 1997). These differences are discussed more fully in the body of the paper.
Movement	Movement is essential to increase intradiscal nutrition, maintain oxygen replenishment, remove waste products, prevent muscle fatigue, and avoid long term compression that may ultimately lead to degeneration (Adams & Hutton, 1983; Graf, Guggenbuhl, & Krueger, 1995; Holm & Nachemson, 1983; Lotz et al, 1998).
Sitter behavior	Studies have shown a preference for reclined, often kyphosed, postures in computer operators engaged in traditional office work or low speed computer inputting, despite discomfort caused by increased neck and shoulder flexion (Graf, Guggenbuhl, & Krueger, 1995; Grandjean 1987; Grandjean, Hunting, & Nishiyama, 1984). Other postures may be assumed because of lack of understanding of the ramifications of such posture on postural health, the nature of the task at hand, chair design, behavior born of habit, or lack of motivation and perceived comfort.

are maintained at right angles. This static form of sitting was recommended from the Victorian era as the traditional ideal posture (Graf, Guggenbuhl, & Krueger, 1995). Seating research between 1948 and 1962 mainly investigated the anatomical and physiological effects of this sitting position, slowly proving it to be a posturally unhealthy concept in that it leads to the adoption of a flexed posture (Andersson, Murphy, Örtengren, & Nachemson, 1979; Bridger, Von Eisenhart-Rothe, & Henneberg, 1989; Keegan, 1953; Schoberth, 1962; Twomey & Taylor, 1987). More recently, radiological studies show that in the right angled sitting position without lumbar support, lumbar lordosis is reduced by 50% compared with standing (Lord, Small, Dinsay, & Watkins, 1997). The reason for this is that sitting in this position creates tension in the hamstring and gluteal muscles, which causes posterior rotation of the pelvis, resulting in a lessening of the sacral horizontal angle and a flattening of the lumbar lordosis (Adams & Hutton, 1980; Andersson, Murphy,

Örtengren, & Nachemson, 1979; Bridger, Von Eisenhart-Rothe, & Henneberg 1989; Keegan, 1953; Schoberth, 1962; Twomey & Taylor 1987), as shown in Figure 1.

By constant activation of the erector spinae muscles during sitting, it is still possible to maintain a lumbar lordosis in the right angled position, without the benefit of a lumbar support. However, such constant muscle activity is unhealthy because it not only creates sustained increases in IVD compressive loading (Andersson, Örtengren, Nachemson, & Elfström, 1974a) but, through more rapid onset of muscle fatigue, it also makes the maintenance of this position untenable for prolonged periods (Floyd & Roberts, 1958). To gain relief from excessive muscle activity, the sitter slides the buttocks forward, thus flexing the lumbar spine, a position that electromyography (EMG) studies show decreases activity in the posterior lumbar muscles when compared with erect sitting (Andersson et al, 1975; Dolan, Adams, & Hutton, 1988). However, there is activation of the neck extensor muscles in order to maintain the cervical spine in neutral (Black, McClure, & Polansky, 1996), posing an unhealthy posture for the cervical spine. This biomechanical configuration explains why a slumped posture is often assumed in a chair with a vertical backrest meeting a horizontal seat.

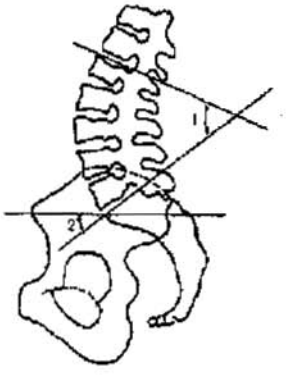
Realizing that the right angled sitting position did not assist spinal postural health, researchers such as Andersson, Örtengren, Nachemson, & Elfström (1974a, b) and Adams and Hutton (1980, 1983, 1985) examined other forms of seated posture. Two schools of thought evolved, one recommending a flexed (kyphosed) lumbar sitting posture and the other recommending a lordosed lumbar sitting posture. Each of these sitting postures was promoted as being beneficial to the IVD, zygapophysial joints, and ligaments and muscles of the lumbar spine, thereby promoting postural spinal health. Often proponents used the same argument to defend their position but with differing conclusions.

Flexed (kyphosed) sitting posture

The lumbar curvature (see Fig. 1, angle 1) may be defined as the angle formed between a line drawn along the superior surface of the sacrum intersecting with a line drawn along the superior surface of L1. This angle averages 60° in the standing lordosed posture, 47° when sitting with lumbar support, and 22° or less when seated unsupported with the spine flexed (Adams & Hutton, 1985, calculated from Andersson, Murphy, Örtengren, & Nachemson, 1979). Postures that flatten the lumbar curve, thereby creating small angles of lumbar curvature, are deemed flexed or kyphosed (Adams & Hutton, 1985).

Effect of the kyphosed sitting posture on IVD pressure

The intradiscal pressure at the level of the third lumbar vertebra is increased by 40% during unsupported slumped sitting with the lumbar spine in kyphosis, when compared with intradiscal pressure in erect standing. This IVD pressure is increased by 85% if the subject leans forward during sitting while the spine remains in kyphosis (Nachemson, 1976). There is evidence suggesting that chronic increases in IVD pressure are responsible for disc pathology (Lotz et al, 1998) and lower back pain (Freemont et al, 1997), which is discussed more fully shortly. The increase in intradiscal pressure in flexed sitting postures is caused by alterations in the center of gravity, load sharing, and ligamentous restraint (Adams, 1995; Andersson, Örtengren, Nachemson, & Elfström, 1974a), in the following manner. In sitting with the hips flexed (beyond 90°) and with a kyphosed lumbar spine, the center of gravity is placed anterior to the ischial tuberosities. Compared to a more erect position more body weight then acts downwards on the anterior surface of the disc, causing deformation of the disc (Andersson et al, 1974a). Deformation of the disc in this way stretches the posterior anulus which creates an increase in the tensile force of the posterior surface of the anulus (Hedman & Fernie,

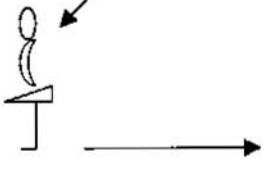


Angles:

1 Lumbar lordosis
2 SHA Sacral Horizontal Angle

Based on Andersson et al (1979)

Sacral horizontal angle (SHA) is the angle between the sacral endplate and a horizontal line intersecting the top of the sacrum at the anterior angle. The greater the SHA the more the lordosis (Schoberth, 1962)



Mean SHA 38 degrees in erect standing. (Andersson et al, 1979)

During standing to sitting, tension occurs in the posterior thigh and gluteal muscles. The sacrum rotates posteriorly, the sacral end plate becomes more horizontal, and SHA evidences a mean of 17.1 degrees in cubist sitting. Thus lordosis is decreased (Andersson et al, 1979)



Cubist position consist of 60 degrees of hip flexion and 30 degrees of pelvic rotation and lumbar flattening

Fig. 1 Changes to the sacral horizontal angle from sitting (cubist position) to standing.

1997; Horst & Brinckmann, 1981) and at the same time wedges the anterior annular surface with a concurrent increase in compressive stress (Andersson et al, 1974a) and load in the anterior anulus (Horst & Brinckmann, 1981; McNally & Adams, 1992). Because of

the concentric arrangement of the lamellae, tensile stresses in the posterior portion of the anulus may be four to five times greater than the applied force, and may be the reason (Nachemson, 1976) that IVD ruptures occur mainly in the posterior anulus (Osti, Vernon-Roberts, Moore, & Fraser, 1992).

Intradiscal pressure is further increased by the effect of the flexed sitting posture on the ligamentous structure. The posterior ligaments resist a considerable amount of the forward bending moment in flexion (Adams, Hutton, & Stott, 1980) and the resultant tensioning, caused by flexion of the lumbar spine (Andersson, Örtengren, Nachemson, & Elfström, 1974a), increases the posterior ligament force approximately three times that of the anterior ligament in extension (Hedman & Fernie, 1997). In extremely flexed positions, tension in the neural arch ligaments can result in a high compressive load on the disc, doubling intradiscal pressure (Adams 1995; Adams, McNally, Chinn, & Dolan, 1994).

The effect of the flexed sitting posture on the zygapophysial joints also contributes to increased intradiscal pressure. In the flexed sitting position, the zygapophysial joints are capable of resisting shear, but not compression forces (Adams, 1995). These compression forces are now borne solely by the IVD and thus intradiscal pressure is further raised (Adams & Hutton, 1980).

Those who advocate the flexed sitting posture acknowledge that an increase in intradiscal pressure occurs in this position, but consider this increase sufficiently small to be ineffective in creating disc damage (Adams & Hutton, 1985; Dolan, Adams, & Hutton, 1988). However, increases in intradiscal pressure ranging from 0.8MPa and 1.49MPa have been demonstrated in upright unsupported sitting (Nachemson & Morris, 1964) and an increase in intradiscal pressure within this range, sustained over a period of hours, has been shown to be deleterious to the IVD of mice, ultimately promoting disc degeneration (Lotz et al, 1998). These authors demonstrated that static compressive loads alter the biological activity of the mouse disc and result in decreased biomechanical performance, irreversible disorganization in the anulus, and cell death. The fluid forced out of mouse discs by a sustained spinal compression of 0.8 MPa over one week decreased nucleus volume and led to a loss of tensile stress in the inner anulus, accompanied by architectural breakdown in the inner and middle anulus. A correlation also was demonstrated between cell death in the nucleus and inner anulus and sustained mechanical loading in the IVD. There is a critical point in loading (1.3 MPa) where sufficient cell death has occurred to prevent the disc recovering when loading is removed. A constant state of compression in the anulus then occurs that alters the biological activity of the disc.

It should be noted that there are several limitations in extrapolating results from mouse studies to humans. Notochordal cells present in the mouse disc are not seen in the adult human; growth plates in the mouse vertebral end plates are not found in adult humans, mice lack laminae, laminar processes, and zygapophysial joints (Lotz et al, 1998), and the disc surface area is obviously smaller. However, the data gathered by Lotz and coauthors demonstrated that IVD degeneration was induced in mouse discs by sustained compressive loading alone, at pressures similar to those achieved in unsupported sitting, and that these degenerative changes were the same as those seen in humans. This evidence provides some support for the theory that a deleterious effect can be triggered when a human disc is consistently compressed above resting loads, such as occurs in the flexed sitting posture.

Effect of kyphosed position on IVD creep and nutrition

In the course of a day's activities (in humans) there is normally a diurnal loss of between 10–20% in disc height, the underlying cause of which is creep (Bogduk & Twomey, 1987; Botsford, Esses, & Ogilvie-Harris, 1994). Whether creep is more substantial in flexed or lordotic sitting postures is a controversial question within the literature. It has generally been concluded from cadaveric motion segment

studies that a sustained flexed sitting posture induces more creep than an erect sitting posture. In the flexed position the IVD is unable to prevent creep because lumbar flexion alters the shape of the disc, stretching and thinning the posterior anulus which, when attenuated, is unable to prevent fluid loss. This effect is compounded by tensioning of the posterior ligamentous structure causing an increase in intradiscal pressure, and unloading of the zygapophysial joints, preventing them from resisting creep (Adams & Hutton, 1983; Adams, McNally, & Dolan, 1996; Magnusson, Pope, & Hansson, 1995; McMillan, Garbutt, & Adams, 1996). Water loss from the nucleus loads the posterior anulus, with the resulting stress concentrations creating the potential for the lamellae to collapse into the nucleus, as well as inducing pain from the posterior anulus and the vertebral body end plate (Adams, McMillan, Green, & Dolan 1996).

An in–vitro study by Hedman and Fernie (1997) was the first to examine the combined response of zygapophysial joints and IVDs to vertical creep deformation and intervertebral force transmission in the flexed and extended postures. These authors found more creep deformation in the extended posture. However, they also reported flexed sitting postures created large increases in the tensile forces in the posterior anulus, which did not occur in extended sitting postures. These potentially damaging forces were due to the significantly increased compressive loads and shear force in the disc in the flexed sitting posture combined with excessively high force escalation in the posterior ligaments as a result of creep-related loading in the flexed sitting posture. A compressive load in the flexed sitting lumbar spine, sustained over six hours, decreases disc height by an average of 2.1 mm (Adams & Hutton, 1983; Adams, McNally, & Dolan 1996; McMillan, Garbutt, & Adams, 1996), narrowing the intervertebral foramen (Adams & Hutton, 1983) and creating the potential to trap nerve structures. Sustained loading of the IVD in the flexed sitting position also leads to a decrease in fluid content in all parts of the disc except the posterior 2 mm of the anulus (McMillan et al, 1996). This loss of fluid creates a severely dehydrated disc (McMillan et al, 1996) and consequently decreases nutrition and accelerates disc degeneration (Handa et al, 1997; Ohshima et al, 1989).

However, a beneficial effect on disc nutrition is one of the arguments used by those favoring flexed sitting postures. In cadaveric motion segment studies Adams and Hutton (1983, 1985) found that cadaveric discs that were continuously loaded to simulate sustained flexed sitting exhibited more creep than those loaded to simulate erect sitting. From this they concluded that flexed sitting positions increase fluid flow. This can aid the nutrition of the lumbar discs. Improved matrix synthesis has been demonstrated with increased fluid flow such as that which occurs in flexed positions, but only when compression is applied and released (Ishihara et al, 1996) as would occur in moving from a flexed sitting position to a lordosed one, or from sitting to standing. The initial movement into flexion may increase fluid flow, but sustained kyphotic positions in sitting cause creep which produces an extremely dehydrated disc (McMillan, Garbutt, & Adams, 1996). Therefore, the argument of Adams and Hutton (1983, 1985) may be flawed as well as functionally invalid, if one considers the proposition that movement also is necessary to IVD nutrition.

Effect of kyphosed position on the zygapophysial joints

Further arguments offered by those advocating the flexed sitting posture involve the zygapophysial joints. It is argued that in the flexed sitting position, compression loading on the zygapophysial joints is relieved, while in the lordotic posture zygapophysial joint loading leads to impingement and articular damage, particularly in the presence of IVD degeneration (Adams & Hutton, 1980, 1985). However, this argument fails to consider that the intra-articular surfaces of the zygapophysial joints show two areas of impaction, one the result of anterior sagittal translation during flexion and the other the result of the rocking

motion of extension (Bogduk, 1997). Therefore, both the flexed and extended sitting positions are a potential cause of zygapophysial damage.

Continuing their argument in favor of flexed sitting postures, Adams and Hutton (1980, 1985) propose that the zygapophysial joints undergo degenerative change because of compressive loads sustained in lordosed sitting positions and this degeneration is a subsequent source of pain. However, if the work of Lotz et al (1998) is applicable to humans, then the sustained increase in intradiscal pressure that occurs in flexed sitting postures promotes disc degeneration, which also has been shown to be a source of pain. In healthy individuals, only the outer third of the anulus is innervated (Gronblad, Weinstein, & Santavirta, 1991; McCarthy, Carruthers, Martin, & Petts, 1991), whereas in the presence of IVD degeneration an ingrowth of pain-sensitive nerve fibers as far as the nucleus of the IVD has been found (Freemont et al, 1997). This finding suggests that both zygapophysial joint and disc degeneration may precipitate low back pain. Therefore, the rationale of advocating flexed sitting posture to prevent pain by relieving potentially damaging loads on the zygapophysial joints may not be substantiated, in that such loads are transferred to the discs, thereby increasing pressure and ultimately leading to disc degeneration, which is also capable of provoking pain.

Another argument advanced by those favoring flexed sitting postures is that cultures that adopt a flexed floor sitting posture show less evidence of back pain and disc degeneration. Because of this it has been suggested a general return to such floor sitting positions as in childhood may be beneficial (Adams & Hutton, 1985; Fahrni, 1975; Gurr, Straker, & Moore, 1998). The radiological study of Fahrni and Trueman (1965) is often cited to support this argument. This study postulates that lumbar lordosis may be implicated in the pathogenesis of lumbar IVD degeneration. The findings are based on radiological evidence that a primitive Indian population whose lifestyle involved sleeping and sitting on the ground exhibited more flattening of the lumbar lordosis and less IVD degeneration than evidenced in a Western population whose occupations were either sedentary or heavy lifting. These conclusions, however, fail to consider occupational, cultural, and environmental factors that may have affected the degree of lumbar IVD degeneration in the Western population. Converse results to Fahrni's (1975) have been found in other radiological studies in a similar Western population (Farfan, Huberdeau, & Dubow, 1972). These researchers observed a correlation between increased IVD degeneration at the lumbosacral level and a lack of lumbar lordosis, concluding that the lumbar lordosis served as a protective mechanism for the lumbosacral region.

Fahrni (1975) also postulates that chair sitting in the right angled position maintains lumbar lordosis and is therefore a major factor in the differences in degeneration and frequency of back pain seen between floor sitting and chair sitting cultures. A study of the tracings from radiographs of various seated positions (Keegan, 1953) shows that lumbar kyphosis is greater when sitting in the right angled position with the lumbar spine unsupported than it is when sitting on the floor crosslegged. Therefore, Fahrni's (1975) argument fails to account for the possibility of a flexed sitting posture when seated in a chair, particularly in the right angled position.

Although not a direct advocate of the flexed sitting position, until the last edition of the back pain series Cailliet (1995) also recommended sitting in the right angled position without lumbar support (1981, p. 131), with the knees higher than the hips (1988). He proposed that this position creates a flat lumbar curve (Cailliet 1988), when in fact it forces the spine into kyphosis. The same position is recommended in *Macnab's Backache* (McCulloch & Transfeldt, 1997, p. 444). It is interesting to note that in 1995, Cailliet advocated maintenance of lumbar lordosis in sitting and had changed the text accordingly, although no reason for this change of emphasis was given.

Finally, those advocating the kyphotic sitting posture argue that lordosis increases the load on the zygapophysial joints (Adams & Hutton 1980, 1985; Williams, 1974), causes a high compressive load on the posterior fibers of the anulus fibrosis where disc degeneration is most prevalent (Adams & Hutton, 1985; Williams, 1974), promotes disc degeneration (Adams & Hutton, 1985; Fahrni, 1975; Williams, 1974), and leads to pressure on the nerve root in the intervertebral canal (Williams, 1974). Conversely, it is argued that maintenance of slight flexion decreases compressive forces on the zygapophysial joints and posterior anulus and increases nutrition, thereby decreasing the degenerative changes that lead to back pain (Adams & Hutton 1980, 1985; Fahrni, 1975; Williams, 1974).

Empirical research has shown that while the flexed sitting posture may decrease posterior anulus compression and unload the zygapophysial joints, it also serves to increase anterior anulus load (Horst & Brinckmann, 1981) and posterior anulus tensile forces. This increased intradiscal load, combined with deformation of the disc and increased ligamentous tensioning, causes a rise in intradiscal pressure, with resultant increase in IVD creep and decrease in nutrition. If such compression is sustained, it may ultimately contribute to disc degeneration and subsequent pain. Further, although the lordosed sitting posture may load the zygapophysial joints, radiological evidence shows that IVD degeneration precedes zygapophysial degeneration (Butler et al, 1990). Degeneration of the IVD then creates mechanical alterations in the loading of the zygapophysial joints and may be responsible for their subsequent degeneration (Butler et al, 1990).

It would appear that sitting postures that lower the intradiscal pressure are more conducive to spinal health than those postures that directly unload the zygapophysial joints at the expense of raised intradiscal pressure. We can therefore conclude that sustained maintenance of the flexed sitting posture is not conducive to postural health.

Lordosed sitting posture

Opposing the proponents of the kyphotic sitting posture are those who promote the maintenance of some degree of lumbar lordosis during sitting. The natural curve in the lumbar spine is known as the lordosis and is the concave curve formed between the superior surface of the first lumbar vertebra (L1) and the superior surface of the sacrum (see Fig. 1, angle 1). This curve varies between individuals and is attributable to a greater height anteriorly in lumbar IVDs than posteriorly, a similar wedge shape in the fifth lumbar vertebra (L5), and an inclination of each of the vertebrae superior to L5 (Bogduk & Twomey, 1987). In upright sitting with a lumbar support of 4 cm, the lumbar lordosis averages 47° (Adams & Hutton, 1985). When the sacral horizontal angle (see Fig. 1, angle 2) is more than 16° the pelvis rotates anteriorly. The downward slope of the superior surface of the sacrum requires a compensatory angulation in the lumbar vertebrae to maintain an upright posture. This is achieved by the wedged shape of both the lumbo-sacral disc and the L5 vertebral body and the inclination of the lumbar vertebral bodies that gradually changes, so that in the erect posture, the L1 vertebra lies vertically above the sacrum (Schoberth, 1962). Those favoring maintenance of lordosis in sitting argue that kyphotic sitting postures increase ligamentous stress, while lordotic postures benefit spinal health by decreasing intradiscal pressure, increasing nutrition, and decreasing disc degeneration (Andersson, Örtengren, Nachemson, & Elfström, 1974a; Hedman & Fernie, 1997; Lotz et al, 1998; McKenzie 1981).

Effect of lordosed position on the lumbar spine structures

Disc pressure measurements taken during in vivo experiments with pressure transducers have demonstrated a reduction in intradiscal pressure in the lordotic sitting position when compared with the kyphosed sitting position (Andersson et al 1975; Nachemson 1966). In cadaveric studies in which the IVD and adjacent

vertebrae were loaded in a simulated lordotic position (i.e., 2° of extension per motion segment), intradiscal pressure was shown to be 36% less than the pressure attained in a simulated kyphotic posture (4–8° of flexion) (Adams, McMillan, Green, & Dolan, 1996; McMillan, Garbutt, & Adams, 1996). Note that the mean and standard deviations have not been reported by these authors, thus the validity of this result is unknown. The comparative reduction in intradiscal pressure during lordosed sitting is the result of lower tensile stresses in the posterior ligaments (Hedman & Fernie, 1997) and a reduction of back extensor activity, provided that adequate lordotic lumbar support is used while sitting in the right angled position (Andersson, Murphy, Örtengren, & Nachemson, 1979). The lower disc pressure also indicates a substantial increase in zygapophysial joint weight-bearing as a result of load sharing between the disc and zygapophysial joints (Adams, McMillan, Green, & Dolan, 1996; Horst & Brinckmann, 1981; McMillan, Garbutt, & Adams, 1996).

It has been proposed that zygapophysial impingement occurs when these joints assist the disc to resist compression resulting from the lordotic sitting posture, particularly in degenerated spines, where the disc space is already narrowed (Adams & Hutton, 1980, 1985; Dunlop, Adams, & Hutton, 1984). It is this increased load on the zygapophysial joints that is often cited (Adams & Hutton, 1980, 1985) as a detrimental effect of lordosed sitting posture. Conversely, however, it also has been shown, following prolonged sitting in a lordosed position, that a balancing of zygapophysial impingement and zygapophysial forces occurs, from which it was calculated that there is a reduction in zygapophysial impingement (Hedman & Fernie, 1997).

Hedman and Fernie (1997) also reported certain benefits of the sustained seated lordosed lumbar position compared with kyphosed postures. They found compressive forces in the disc were reduced, disc shear force was decreased by 75%, and there was a balancing of forces in the zygapophysial joints. In the sustained kyphotic sitting position, however, there was a 9% increase in disc shear force, large increases in the tensile forces around the posterior anulus, and greater, unbalanced anterior forces in the zygapophysial joints. Further, three times the level of force occurred in the posterior ligament in the flexed position than occurred in the anterior ligament in the extended position. Hedman and Fernie (1997, p. 746) referred to results of the aggregate posterior ligament and force escalation that occur from loading in the flexed posture resulting from creep as ''alarming'' in terms of potential damage to the posterior ligament and disc. Thus, it can be concluded that the potential for IVD degeneration and low back pain is increased in the sustained kyphosed sitting position.

Clinical trials also have demonstrated the efficacy of lordosed seated postures in reducing back and, in certain instances, leg pain. In a randomized study that excluded pathologies aggravated by lordotic positions (spondylolisthesis and spinal stenosis), it was demonstrated that 48% of subjects with pain extending below the knee experienced reduction and centralization of pain towards the spine when sitting in the lordosed posture. Only 10% of subjects with leg pain sitting in a kyphotic posture achieved this result (Williams, Hawley, McKenzie, & van Wijmen, 1991). Donelson, Silva, and Murphy (1990) reported similar improvement of back pain in subjects who evidenced intact anulus on positive discograms, and were treated with repeated passive extension exercises as advocated by McKenzie (1981). This clinical finding would indicate that, provided the nucleus has not herniated but is exhibiting internal disc disruption as a result of compression (Bogduk, 1997), lordotic sitting postures reduce discogenic pain more effectively than kyphotic postures.

In summary then, a sustained lordosed sitting posture decreases disc pressure and thereby disc degeneration, exhibits less injurious levels of ligament tension, and although it increases zygapophysial loading, this is not of itself considered hazardous to spinal health. A sustained kyphosed sitting posture, on the other hand, increases intradiscal pressure leading

to increased fluid loss, decreased nutrition, and altered cell synthesis and biomechanics of the disc, appearing to culminate ultimately in disc degeneration that is a cause of low back pain. Therefore, maintenance of the lordosed lumbar spine is less detrimental to seated postural health than maintenance of the kyphosed lumbar spine.

Maintaining lordosis

The question now arises, how is the lordosis best maintained? To eliminate the posterior rotation of the pelvis and concomitant flattening of the lumbar spine, the angle between the hip and the trunk needs to be increased (Keegan, 1953; Schoberth 1962), thus reducing hip flexion. In the extreme case, a greater reduction in hip flexion may be achieved by assuming a semi-lying position. This position is advocated by Cranz (1998) as part of her ideal workplace for office and computer tasks. However, it should be acknowledged that while such a position as advocated by Cranz may reduce intradiscal pressure, it may not be functional and could be hazardous to postural health, unless suitable support is afforded the spine in its entirety and the equipment (e.g., the document holder, writing surface, monitor, keyboard, and armrests) is positioned to prevent unnecessary muscle activity of the neck and arms. Until these design issues are addressed, a more conservative approach to decreasing hip flexion and maintaining postural health may be achieved by either reclining the backrest or tilting the seat incline forward. The effects of reclined and forward tilt seating may be seen in Tables 2 and 3.

In summary, for tasks not involving a forward posture, it has been shown when a lumbar support of 4 cm depth is used in conjunction with a reclined backrest of 130°, lumbar lordosis is restored close to that of standing and intradiscal pressure is decreased approximately 65% compared with erect sitting without lumbar support (Andersson et al, 1975; Andersson, Murphy, Örtengren, & Nachemson, 1979). Therefore, the combination of reclined seating and lumbar support assists in providing seated postural health for those engaged in conversational or relaxation activities.

However, reclined seating is not always the optimal position for those engaged in traditional desk work or intensive keyboard tasks, where the task commonly necessitates a forward orientation of the body. Space constraints preclude an in-depth description of the effect of reclined seating on the cervical and glenohumeral joints. Briefly, if the trunk is reclined beyond 110°, cervical and gleno-humeral flexion must occur in order to view and access the desktop (Chaffin & Andersson, 1991; Hochanadel, 1995). In keyboard tasks, flexion of the cervical spine in reclined sitting

Table 2
Effect of backrest recline 100–110

Effect of recline	Research findings
Pressure in the lumbar IVD is reduced	At a recline of 105° 4.4% of body weight is transmitted to the backrest (Swearingen, Wheelwright, & Garner, 1962), decreasing disc pressure by 0.15MPa (Andersson, Örtengren, Nachemson, & Elfström, 1974a). At a recline of 130° intradiscal pressure is reduced by approximately 50% (Andersson et al, 1975).
Creep in the lumbar IVD is reduced	Reduced disc pressure reduces fluid loss. More significant reduction of fluid loss is seen at 110° recline than at 105° (Magnusson, Hansson, & Pope, 1994).
Static muscle activity is reduced	Reduced posterior muscle activity of the cervical, thoracic, and lumbar spines reduces lumbar IVD pressure (Andersson, Murphy, Örtengren, & Nachemson, 1979; Andersson, Örtengren, Nachemson, & Elfström, 1974a).
Lumbar lordosis is not restored	The insertion of a well designed convex lumbar support is necessary to restore lumbar lordosis (Andersson, Murphy, Örtengren, & Nachemson, 1979).

Table 3
Effects of forward tilt seating

Effect of forward tilt	Research findings
Restoration of lumbar lordosis	Restoration of the lumbar lordosis close to that of standing was shown by radiological studies of one subject seated on a stool with a hip/trunk angle of 135° (Keegan, 1953, p. 600). Radiological studies by Colombini et al, (1986) showed with a larger sample (n = 10) that subjects sitting on a stool with a slope of 20° evidenced less reversal of lumbar lordosis compared with other seated positions (Colombini et al, 1986). These findings were corroborated by Bridger, von Eisenharte-Rothe, and Henneberg (1989) in a repeated measure, random order study.
Relief of hamstring and gluteal tension	Creates forward rotation of the pelvis and restoration of lumbar lordosis to slightly less than seen in the standing position (Bendix & Biering-Sorenson, 1983; Bridger, von Eisenharte-Rothe, & Henneberg, 1989; Bridger, Wilkinson, & van Houweninge, 1989; Keegan, 1953).
IVD pressure and lumbar muscle activity are reduced	A maximal lowering of biomechanical stress was shown by Colombini et al, 1986 with subjects sitting on a forward tilt seat of 20°, compared with other forms of seating.
Discomfort with the use of a conventional seat with a forward slope of 15°	Problems of forward slip, dragging of clothing, and increased weight on the feet (Bendix & Biering-Sorenson, 1983; Bendix, Winkel, & Flemming, 1985; Graf, Guggenbuhl, & Krueger, 1993).
Discomfort with the use of a sit kneel seat	Difficulty with ingress/egress lack of movement for chair and occupant, discomfort on shins (Drury & Francher, 1985), lack of backrest creating increased erector spinae activity, and kyphosis with use past 10 seconds (Lander, Korbon, DeGood, & Rowlingson, 1987).

may be eliminated by elevating the monitor. However, static glenohumeral flexion remains necessary to reach the keyboard. Static muscle activity in the neck flexors and the trapezii has been shown to be detrimental to postural health (Harms-Ringdahl & Schuldt, 1990; Villaneuva et al, 1997). Muscle fatigue is hastened as a consequence (Harms-Ringdahl & Schuldt, 1990) and there is a tendency to relax the shoulder musculature by resting the forearms on the desk, thus potentially risking compression of the soft tissues of the carpal tunnel. To avoid such unhealthy postures and to restore lumbar lordosis, the concept of forward tilt seating has been recommended (Burandt, 1969; Carlsöö, 1963, Mandal, 1975, 1976; Rosemeyer, 1972; Staffel, 1884). This position is likened to the erect posture assumed by horse riders, the degree of hip flexion allowing maintenance of the lumbar lordosis despite the lack of backrest (Mandal 1975, 1976). Lack of comfort is reflected by poor subject preference studies comparing the kneel-sit chair with more conventional seating (Bendix, 1988; Bendix, Winkel, & Flemming, 1985; Drury & Francher, 1985; Lander, Korbon, DeGood, & Rowlingson, 1987). However, current designs of forward tilt chairs, including kneel-sit, have addressed these problems (see Table 4), now rendering forward tilt seating a viable and posturally healthy manner of sitting.

Therefore, it can be concluded that both reclined and forward tilt seating have biomechanical advantages for postural health. Subjective and postural health research has shown the choice between the two angulations to be task dependent. On balance, the research tends to indicate that more researchers are now agreeing that for forward oriented tasks such as desk or computer-intensive work, a forward tilt of the seat with backrest is desirable (Dainoff, 1994; Festervoll, 1994; Pheasant, 1991; Shiraishi & Ueno, 1994). A design that has addressed problems of forward slip, pressure, and back support should be sought. For tasks not involving a forward posture, backrest recline is desirable. However, reclining the backrest does not in itself restore lumbar lordosis. Further, many chairs currently used in offices either have no provision for forward tilt, or are designed with a maximal forward tilt of 5°. This degree of forward tilt is based on research demonstrating that at this inclination, forward slip is overcome and the need to brace the legs is negated (Bendix & Biering-Sorenson, 1983). In such forward tilt seating, right angled seating

Table 4
Improvements in forward tilt seating design

Type of seat	Improvement in design
Kneel-sit (Stokke and Equipoise chairs)	Unimpeded ingress/egress, wider tibial support decreasing pressure, backrest synchronised to both recline and forward tilt postures, both of which are available with change of weight.
Conventional seat with sloping seat pan	a) A seat pan horizontal under the ischial tuberosities and tilted forward under the thighs improves the lumbar lordosis by 4° (Corlett & Ecklund, 1984; Graf, Guggenbuhl, & Krueger, 1993). b) Combining pelvic and thigh tilt to 15° improves lordosis by 13° (Bendix & Biering-Sorenson, 1983). This was corroborated by repeated measure, random order study (Bridger, von Eisenharte-Rothe, & Henneberg, 1989). c) The study by Wu, Miyamoto, and Noro (1998) investigating the use of a pelvic support shows some confusion in documentation. These authors concluded that a 4° forward sloped seatpan enhanced by the addition of a 10° forward slope under the pelvis, improves stability and comfort while decreasing forward slip.

(Åkerblom, 1948; Keegan, 1953; Schoberth, 1962), and in reclined seating (Andersson, Murphy, Örtengren, & Nachemson, 1979), the addition of a lumbar support in the form of a well designed convex pad, allows passive maintenance of lumbar lordosis.

Lumbar support

The literature contains various views regarding the use, placement, and depth of lumbar supports. These are summarized in Table 5.

Andersson and colleagues (Andersson, Örtengren, Nachemson, & Elfström, 1974b; Andersson et al, 1975) have shown if the lumbar support is placed at the level of the sacrum it will have the effect of forcing the buttocks forward, creating lumbar kyphosis. Further, it is our opinion that as backrest recline or seatpan incline decreases pelvic rotation, placing the lumbar support at the pelvic rim to prevent rotation becomes unnecessary in reclined or forward tilt seating. We agree with McKenzie (1981), who argues that, provided the apex of the lumbar support is placed so as to maintain the lordosis but not at end of range extension, and such a placement creates a pain-free and relaxed position, this will be the optimal location for lumbar support of the seated individual.

We further suggest that a set depth of the lumbar support does not account for differences in anatomical configuration. Less than half the population exhibit a ''normal'' lumbar lordosis (Branton, 1984; Schoberth 1962, 1969). A recommended depth does not account for the greater natural lordosis in females when sitting, or for the differences in lumbar curvature for both sexes dependent on seat slope (Bridger, Von Eisenhart-Rothe, & Henneberg, 1989). Postural health is better served if the depth of the lumbar support is appropriate to the degree of individual lordosis. This avoids holding the lumbar spine in sustained hyperlordosis, which causes sufficient vertical compression of the zygapophysial joints that the inferior process of the joint may impinge on the lamina of the vertebra below (Bogduk & Twomey, 1987), causing pain. Depth of lumbar support therefore needs to be adjustable to accommodate individual lordosis. McKenzie (1981) suggests finding the required depth of the lumbar support by asking the person to move from maximal slumped to maximal erect sitting several times, and then relax the back 10% from the maximal position. In this way the patient not only is taught the difference between kyphosis and lordosis, but also learns the optimal lumbar sitting position.

In a well designed study examining subject preference, Coleman, Hull, and Ellitt (1998) demonstrated that subjects more readily adjust the depth than the height of the lumbar support, a factor due possibly to ease. However, the lumbar support should be adjustable in

Table 5
Lumbar support placement and depth

Support placement or depth	Placements/depths advocated by various authors
Placement of the apex of the lumbar support	Alternative positions advocated include: a) Low lumbar spine (Branton, 1969; Keegan, 1953, 1962), b) Mid lumbar spine (Sauter & Arndt, 1984), c) Between L2 and L5 (Floyd & Roberts, 1958), d) L3–L4 (McKenzie, 1981; Williams, Hawley, McKenzie, & van Wijmen, 1991), e) Pelvic rim to prevent rotation of the pelvis in either the forward or backward tilted seat position (Grandjean, 1973; Schoberth, 1969; Stumpf, Chadwick, & Dowell, 1995) f) Low lumbar spine in those with back pain and the high lumbar spine in obese subjects, but no more than 20cm above the compressed seat (Coleman, Hull, & Ellitt, 1998).
Depth	Alternative depths advocated: a) The ideal depth to reduce disc pressure and restore lumbar lordosis close to that of standing in 4cm (Andersson, Murphy, Örtengren, & Nachemson 1979). b) The ideal depth in sitting is that which is required to maintain the lumbar spine 10% off maximal lordosis (McKenzie, 1981).

both the vertical and horizontal dimensions to ensure correct individual fit and to maximize postural health benefits. Provision of such adjustment accommodates the broad spectrum of physiques (Coleman, Hull, & Ellitt, 1998) and the multi-task nature of sedentary occupations. The lumbar support has been shown to move upwards by 4.5 cm when the backrest is reclined from 90° to 105° (Andersson, Murphy, Örtengren, & Nachemson, 1979). Therefore, the lumbar support must be adjusted in a caudal direction when the backrest is reclined and in a cephalad direction when the backrest is inclined forward. The latter adjustment is essential to prevent the lumbar support contacting the sacrum and buttocks.

THE ROLE OF THE CHAIR IN HEALTHY SITTING

Reclined seating, forward inclined seating, and enhancing the lumbar lordosis are all functions of chair design. Office workers today require a chair that can maintain all these healthy postures to accommodate the multi-task nature of their duties. It is important, therefore, to provide seating with adjustments that are readily reached and manipulated, respond immediately, accommodate varying body shapes and sizes, and are easily understood (Helander, Zhang, & Michel, 1995; Herman Miller, Inc., 1994). The most frequently used adjustment is seat height (Coleman, Hull, & Ellitt, 1998; Helander, Zhang, & Michel, 1995; Hochandel 1995). The reticence to make use of other adjustments may be reflective of the bewildering array of adjustments on modern chairs, and the lack of training and reference to user manuals (Helander, Zhang, & Michel, 1995). It also may reflect user ignorance regarding the impact of seated posture on health. This ignorance may be overcome by stressing that the chair's role in healthy sitting posture necessitates the use of adjustments that have been provided to ensure maintenance of lumbar lordosis while allowing alternating postures between reclined and forward tilt seating, thus avoiding one sustained posture. It is equally important to ensure that the subject chooses a chair that addresses the task at hand and the uses to which the seat will be put. This advice will avoid the necessity for extremes of posture by fitting the chair to the occupant and the work space. When advising on chair choice, therefore, not only anatomy and physiology should be considered, but also the workplace, the tasks, the behavior, and the vast variety of physical differences of the individual (Graf,

Guggenbuhl, & Krueger, 1993, 1995; Hermann Miller, Inc., 1994; Waldron & Edling, 1997).

CONCLUSION

Cadaveric studies demonstrate that disc degeneration is associated with sedentary work (Videman, Nurminen, & Troup, 1990) and epidemiological studies show that sedentary occupations are a cause of low back pain (Kelsey, 1975; Magora, 1972). Poor seated posture, sitting behavior and seat design are all thought to be contributing factors to this pain. Poor seated posture may be addressed by alerting the sitter to the potential damage caused to the discs, joints and soft tissues by sustained kyphotic postures, and to the benefits to postural health arising from maintenance of lordosis, interspersed with movement. Rhythmic movement of the lumbar spine through lordosis to kyphosis is beneficial to disc nutrition, provided the kyphosed position is not sustained. Movement into standing from sitting also should be encouraged to relieve disc compression. It should be emphasised that no one posture is ideal, and that healthy sitting requires a chair that allows posture change and is easily adjustable between reclined and forward tilt, dependent on task demands. Finally, it is essential that the sitter understands that pain arising from poor seated posture is insidious, and that healthy backs may not register pain until irreversible problems (e.g., IVD degeneration) have occurred. For this reason, the teaching of healthy seated posture should be seen as both preventive and therapeutic.

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